1.

sign : +

exp : 24 − 127 = −103 $\Rightarrow 1 \times 2^{-103}$

$= 9.86 \times 10^{-32}$ #

2. largest positive : $0\ \underline{11111110}\ \underbrace{11\cdots1}_{23} = 1.111\cdots1 \times 2^{127}$ or 3.4×10^{38}

smallest positive : $0\ \underline{00000001}\ \underbrace{00\cdots0}_{23} = 1 \times 2^{-126}$ or 1.2×10^{-38}

smallest number : $1\ \underline{11111110}\ \underbrace{111\cdots1}_{23} = -1.11\cdots1 \times 2^{127}$ or -3.4×10^{38}

3.

Guard bit:有效位數後一位
Round bit:有效位數後第二位
Sticky bit:除了Guard bit、Round bit外，後方不全為零則sticky bit=1

44454400

0 10001000 100 01010100010000000000

$= 1.100\ 01010100010000000000 \times 2^{9}$

451FE980

0 10001010 00111111110100110000000

$= 1.00111111110100110000000 \times 2^{11}$

$0.01\ 100\ 01010100010 \times 2^{11}$
$1.00111\ 11111010011 \times 2^{11}$

$1.101\underline{00010011101}01 \times 2^{11}$
GR S=1

$= 1.1 \times 2^{11}$ #

4. $185 - 122 = 63$ neither

5. (1) 9 (2) 19 (3) 3 (4) 19 (5) 18

(6) 4 (7) 24 (8) 9 (9) 8 (10) 706

6. $119.25 = 1110111\ 01$

$= (-1)^0 \times (1 + 0.11011101) \times 2^6$

$exp = 6 + 127 = 133$

0 10000101 11011101 00....0

sign exponent fraction 15

#